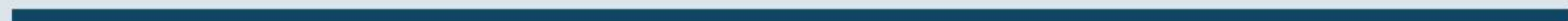


Automated Parcel Damage Detection System for E-Commerce Fulfillment Centres



LARSEN & TOUBRO



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Introduction

The rapid growth of e-commerce has increased the need for efficient parcel inspection in fulfillment centers. Manual inspection is time-consuming and may lead to errors when handling large volumes of packages. This project proposes an Automated Parcel Damage Detection System using YOLO-based Computer Vision technology to identify damaged and non-damaged parcels automatically. The system analyzes parcel images, predicts their condition, and generates structured outputs for monitoring. By automating the inspection process, it improves accuracy, reduces processing time, and enhances overall operational efficiency in logistics and e-commerce warehouses.

Problem Statement

In e-commerce fulfillment centers, thousands of parcels are handled and shipped daily. Manual inspection of packages is time-consuming, labor-intensive, and prone to human errors. Damaged parcels may go unnoticed, leading to customer dissatisfaction, product returns, and financial losses. Therefore, an automated system is required to accurately detect parcel damage and improve the efficiency of the inspection process.

Objectives

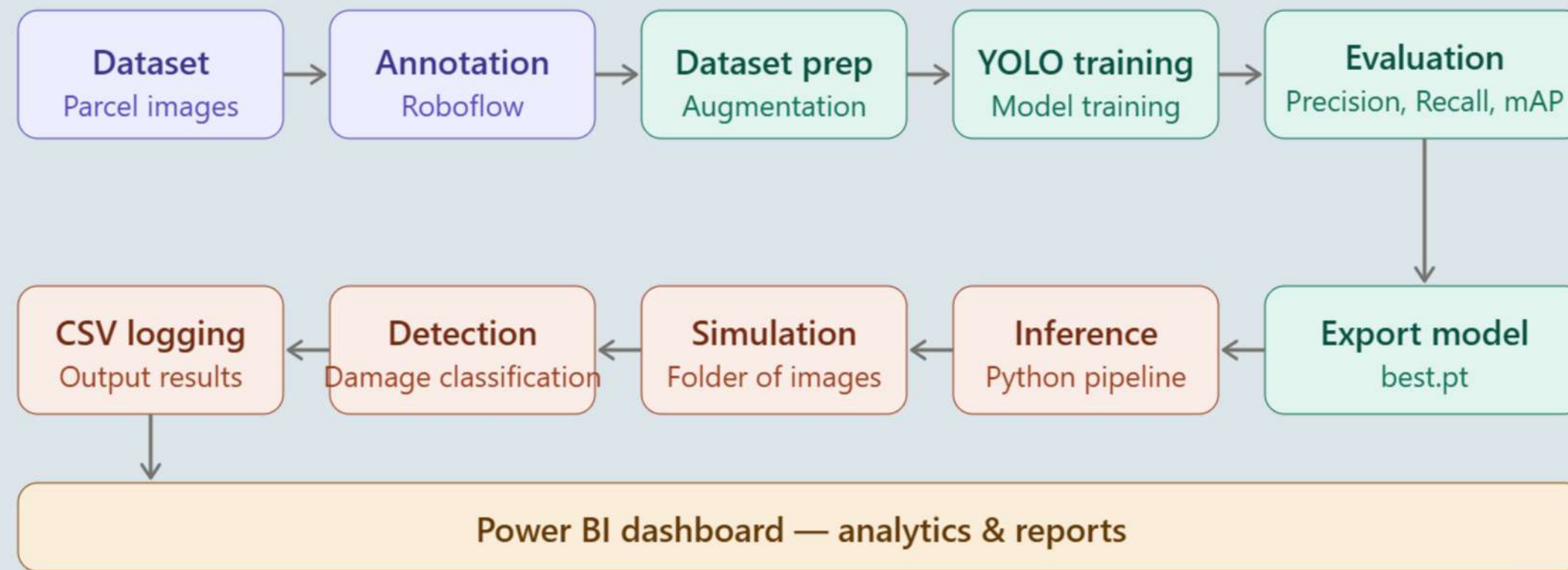
- To develop an AI-based system for automatic parcel damage detection.
- To classify parcels into different categories such as Box, Package, Open Package, and Box Broken.
- To reduce manual inspection effort and improve accuracy.
- To generate structured detection results using a Python inference pipeline.
- To visualize parcel inspection data through an interactive Power BI dashboard.
- To improve operational efficiency in e-commerce fulfillment centers.

Challenges

- Collecting a balanced dataset with sufficient damaged and non-damaged parcel images.
- Accurately annotating parcel images for effective model training.
- Handling variations in parcel size, shape, lighting conditions, and camera angles.
- Distinguishing between similar parcel conditions such as Open Package and Box Broken.
- Ensuring high detection accuracy while minimizing false predictions.
- Integrating model predictions with a Power BI dashboard for meaningful analytics.

System Architecture / Methodology

The proposed system follows a structured workflow for parcel damage detection. Parcel images are collected and annotated using Roboflow, followed by dataset preprocessing and augmentation. A YOLO model is trained using the prepared dataset and evaluated for accuracy. The trained model performs inference on new parcel images and generates predictions. The results are stored in a CSV file and visualized through a Power BI dashboard for analysis and monitoring.



Dataset Collection and Augmentation

The parcel image dataset was collected and annotated using Roboflow for training the damage detection model. The dataset consists of four classes: Box, Package, Open_Package, and Box_Broken. **Initially, the dataset contained approximately 1,200 annotated images.** To improve model performance and increase data diversity, augmentation techniques such as **Auto-Orientation, Resizing (640×640), Brightness Adjustment (-15% to +15%), and Exposure Variation (-15% to +15%)** were applied. **After augmentation, the dataset size increased to 2,879 images,** providing more variations of parcel conditions and helping the model generalize better to real-world scenarios.

Model Performance Highlights

mAP@50: 95.2% → Indicates excellent object detection accuracy.

Precision: 96.4% → Very low false-positive detections.

Recall: 90.2% → Successfully detects most damaged parcels.

F1-Score: 93.2% → Demonstrates a strong balance between precision and recall

Training Dataset Summary

Original Images: 1,200

Images After Augmentation: 2,879

Number of Classes: 4

Training Images: 2,519 (87%)

Validation Images: 240 (8%)

Test Images: 120 (4%)

Model: YOLOv11

Metrics ?

Valid Set

mAP@50
95.2%

Precision
96.4%

Recall
90.2%

External ?

F1
93.2%

YOLOv11 Model Training

Versions

2026-06-10 9:54pm

v1

2879

Small

COCOs

Swetha Mohan A

v1

2026-06-10 9:54pm

Generated on Jun 10, 2026 by Swetha Mohan A

Download Dataset

Edit

Package Detection 1

View Model

Model URL:
package-detection-e1ssd-

Checkpoint
COCOs

Metrics

Dataset

mAP@50

95.2%

Precision

96.4%

Recall

90.2%

F1

93.2%

External

Updated On:
Jun 10, 2026, 10:38 PM

Model Type:
YOLOv11 Object Detection (Small)

2879 Total Images

View All Images

Dataset Split

TRAIN SET

87%

2519 Images

VALID SET

8%

240 Images

TEST SET

4%

120 Images

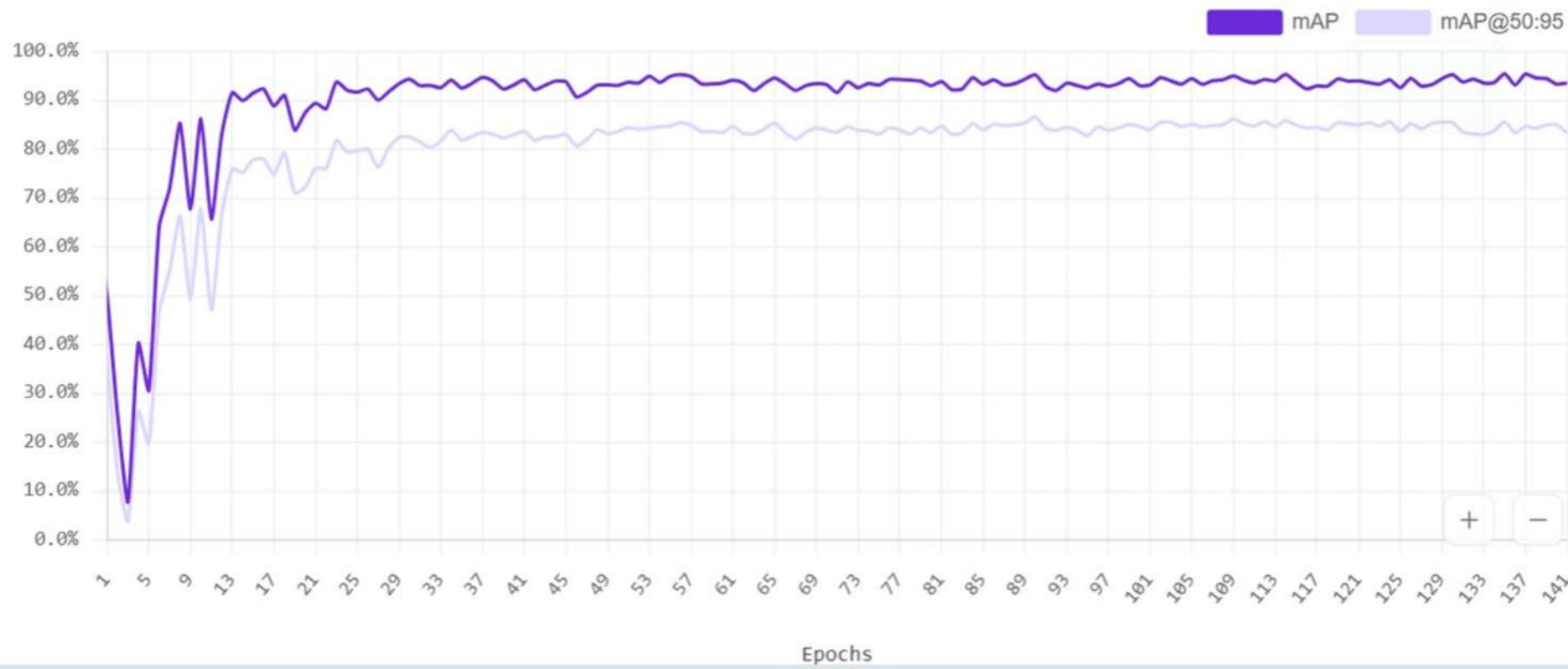
Preprocessing

Auto-Orient: Applied
Resize: Stretch to 640x640

Augmentations

Outputs per training example: 3
Brightness: Between -15% and +15%
Exposure: Between -15% and +15%

Model Performance



mAP Graph

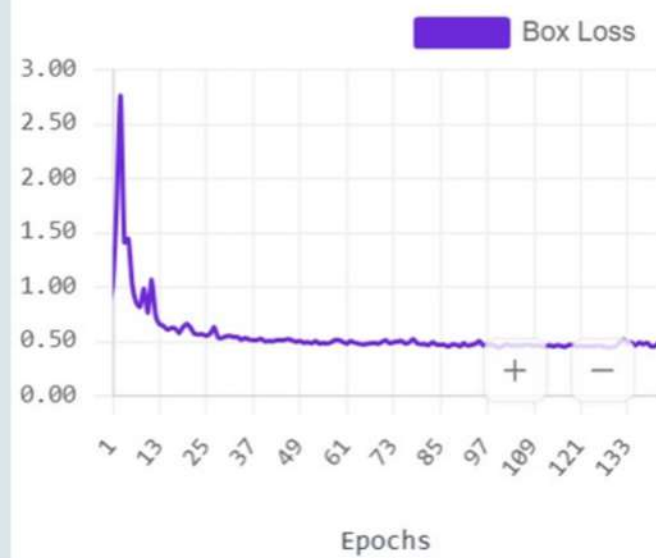
mAP increased steadily during training.

Final mAP@50 reached 95.2%.

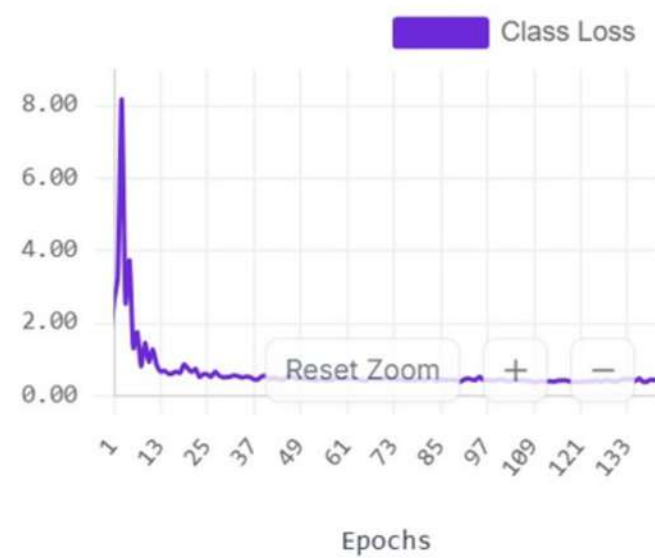
Model achieved high detection accuracy.

Performance stabilized in later epochs.

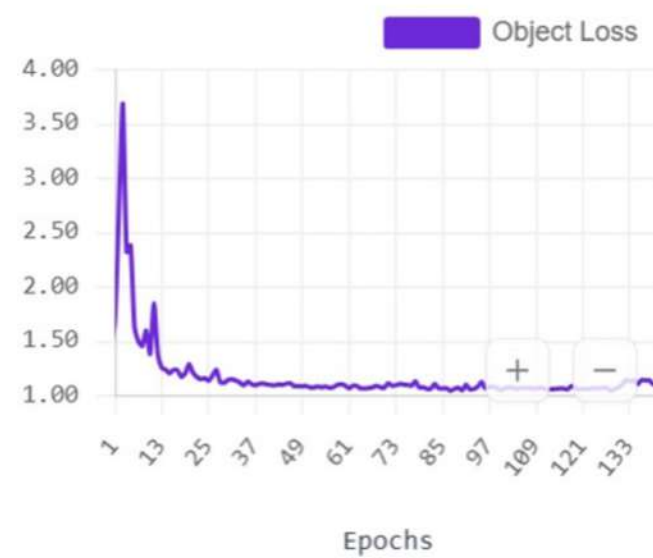
Box Loss



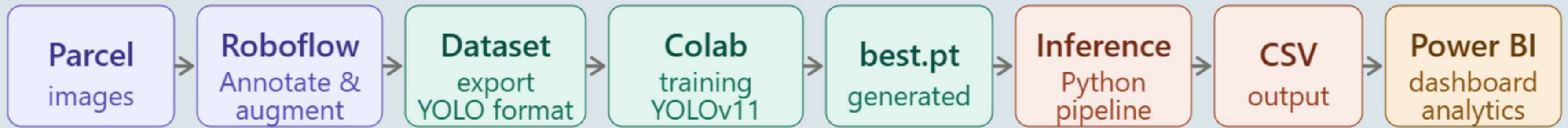
Class Loss



Object Loss



Workflow



Model Training Strategy

Initially, the dataset was uploaded to Roboflow for annotation, preprocessing, augmentation, and model configuration. Roboflow was also used to prepare the YOLOv11 training environment and evaluate dataset quality. However, downloading the trained model weights required a premium upgrade. Therefore, the prepared dataset was exported and the final YOLOv11 model training was performed in Google Colab . After successful training, the best-performing model weights (best.pt) were generated and downloaded for deployment in the Python inference pipeline.

```
from google.colab import files  
  
files.download('/content/runs/detect/train/weights/best.pt')
```

Python Inference Pipeline

After successfully training the YOLOv11 model in Google Colab, the generated model weights file (best.pt) was downloaded and deployed in a local Python environment. A Python-based inference pipeline was developed to automate the parcel inspection process. The pipeline loads the trained YOLOv11 model using the Ultralytics framework and processes parcel images stored in a local folder. Each image is analyzed to detect parcel objects, classify parcel conditions, and generate confidence scores for the predictions. The system performs batch processing of multiple images automatically without requiring manual intervention, making it suitable for large-scale parcel inspection applications. The generated prediction results are then prepared for reporting, visualization, and further analytics.

OUTPUT IN PYTHON INFERENCE

```
=====
Inference Completed Successfully
=====
Total Detections : 146

Class Counts:
Class
Package          43
Open_package     35
Box              34
Box_broken       34
Name: count, dtype: int64

CSV Saved As: detections.csv
```

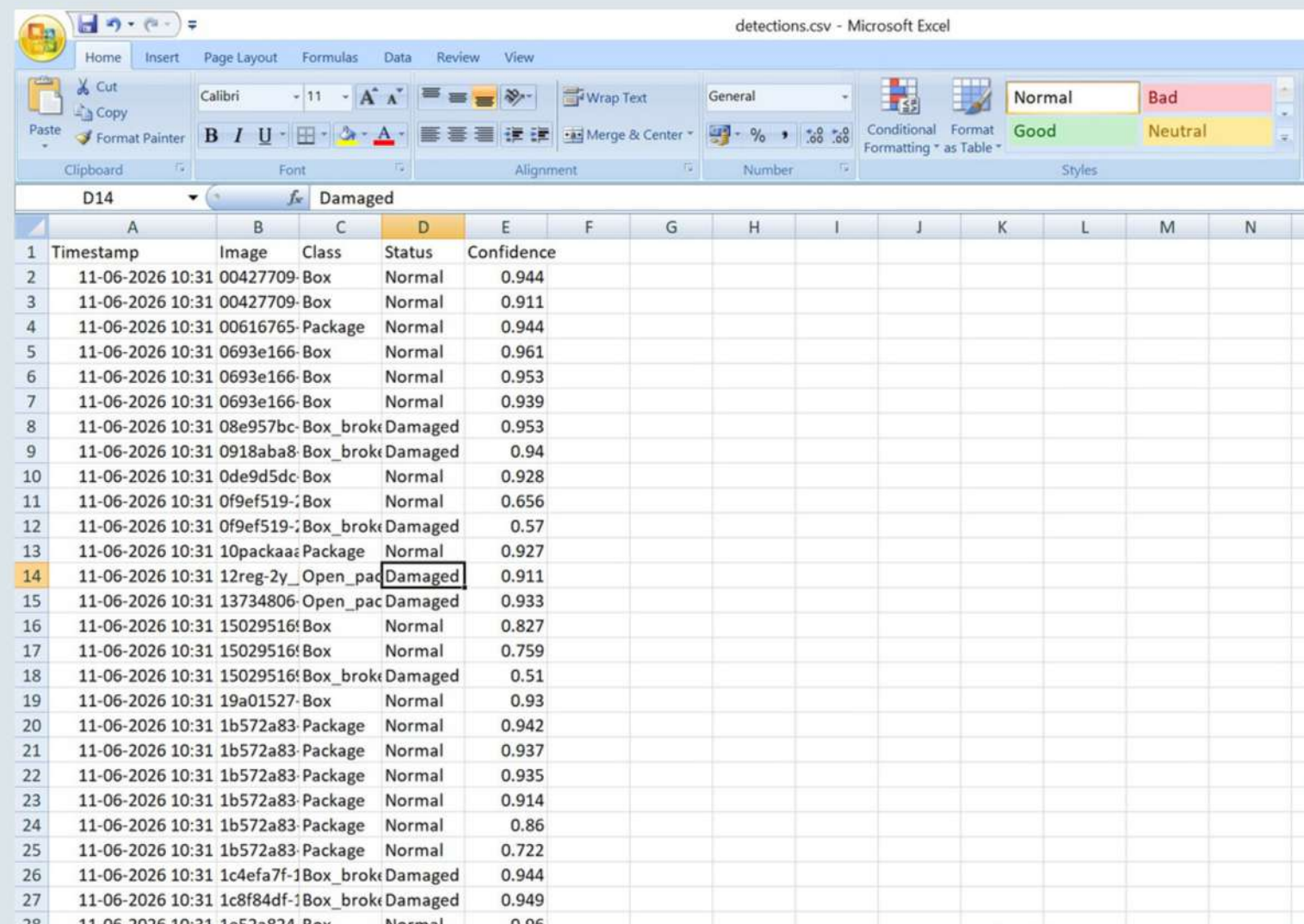
CSV Output Generation

Overview

After the Python inference pipeline processes the parcel images, the prediction results are automatically stored in a CSV (Excel-compatible) file.

The CSV file acts as a bridge between the AI model and the Power BI dashboard.

Each detected parcel is recorded with its corresponding prediction details.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Timestamp	Image	Class	Status	Confidence									
2	11-06-2026 10:31	00427709-Box	Normal	0.944										
3	11-06-2026 10:31	00427709-Box	Normal	0.911										
4	11-06-2026 10:31	00616765-Package	Normal	0.944										
5	11-06-2026 10:31	0693e166-Box	Normal	0.961										
6	11-06-2026 10:31	0693e166-Box	Normal	0.953										
7	11-06-2026 10:31	0693e166-Box	Normal	0.939										
8	11-06-2026 10:31	08e957bc-Box_brok	Damaged	0.953										
9	11-06-2026 10:31	0918aba8-Box_brok	Damaged	0.94										
10	11-06-2026 10:31	0de9d5dc-Box	Normal	0.928										
11	11-06-2026 10:31	0f9ef519-Box	Normal	0.656										
12	11-06-2026 10:31	0f9ef519-Box_brok	Damaged	0.57										
13	11-06-2026 10:31	10packaa-Package	Normal	0.927										
14	11-06-2026 10:31	12reg-2y_Open_pac	Damaged	0.911										
15	11-06-2026 10:31	13734806-Open_pac	Damaged	0.933										
16	11-06-2026 10:31	15029516-Box	Normal	0.827										
17	11-06-2026 10:31	15029516-Box	Normal	0.759										
18	11-06-2026 10:31	15029516-Box_brok	Damaged	0.51										
19	11-06-2026 10:31	19a01527-Box	Normal	0.93										
20	11-06-2026 10:31	1b572a83-Package	Normal	0.942										
21	11-06-2026 10:31	1b572a83-Package	Normal	0.937										
22	11-06-2026 10:31	1b572a83-Package	Normal	0.935										
23	11-06-2026 10:31	1b572a83-Package	Normal	0.914										
24	11-06-2026 10:31	1b572a83-Package	Normal	0.86										
25	11-06-2026 10:31	1b572a83-Package	Normal	0.722										
26	11-06-2026 10:31	1c4efa7f-1Box_brok	Damaged	0.944										
27	11-06-2026 10:31	1c8f84df-1Box_brok	Damaged	0.949										
28	11-06-2026 10:31	1e52a874-Box	Normal	0.96										

CSV Output Results

Prediction results are automatically saved in detections.csv.

Each record contains Timestamp, Image Name, Class, Status, and Confidence Score.

Parcel status is classified as Normal or Damaged.

Power BI Dashboard Integration

Parcel Inspection Log

Timestamp	Image	Class	Status	Average of Confidence
1-06-2026 10:31:54	00427709-156Boooooox_jpg.rf.73f6361f444942f376ba4157b3eeea49.jpg	Box	Normal	0.93
1-06-2026 10:31:54	00616765-67Packageeees_jpg.rf.63a4a4f0a098255fb029a8c54e27a0c0.jpg	Package	Normal	0.94
1-06-2026 10:31:54	0693e166-139Boooooox_jpg.rf.e1c4bd2ae3c8485e6d53fd500f761a88.jpg	Box	Normal	0.95
1-06-2026 10:31:54	08e957bc-145Open_booox_jpg.rf.a9070e1919abc072135d29121ad27bbd.jpg	Box_broken	Damaged	0.95
1-06-2026 10:31:54	0918aba8-93Open_booox_jpg.rf.14cc6f467553c38f8835d76df106ce93.jpg	Box_broken	Damaged	0.94
1-06-2026 10:31:54	0de9d5dc-194Boooooox_jpg.rf.d04a15c00825dddc23ed4c0bd7f78ba.jpg	Box	Normal	0.93
1-06-2026 10:31:54	0f9ef519-210Open_booox_jpg.rf.318b585fe7311d6f5bddfa0cf4510e23.jpg	Box	Normal	0.66
1-06-2026 10:31:54	0f9ef519-210Open_booox_jpg.rf.318b585fe7311d6f5bddfa0cf4510e23.jpg	Box_broken	Damaged	0.57
1-06-2026 10:31:54	10packaaaageeees_jpg.rf.0b7b69a207cbcd1dbc44fc579a31bfc9.jpg	Package	Normal	0.93
1-06-2026 10:31:55	12reg-2y_jpg.rf.c328960715e4aaafd9ae1822007253d1.jpg	Open_package	Damaged	0.91
1-06-2026 10:31:55	13734806-167Opeen_Packages_jpg.rf.77aaf4f76123c91fe4dbbe23da50f50a.jpg	Open_package	Damaged	0.93
1-06-2026 10:31:55	1502051604_26b680ba1b24cd02d560c6b0b7cb122a.jpg.rf.51dccc220a00d0a0c0207db7020ff706.jpg	Box	Normal	0.70

Status

- ☐ Damaged
- ☐ Normal

Class

- ☐ Box
- ☐ Box_broken
- ☐ Open_package

Automated Parcel Damage Detection Dashboard

146

Total Parcels Processed

69

Rejected Parcels

77

Accepted Parcels

47.26

Damage Rate %

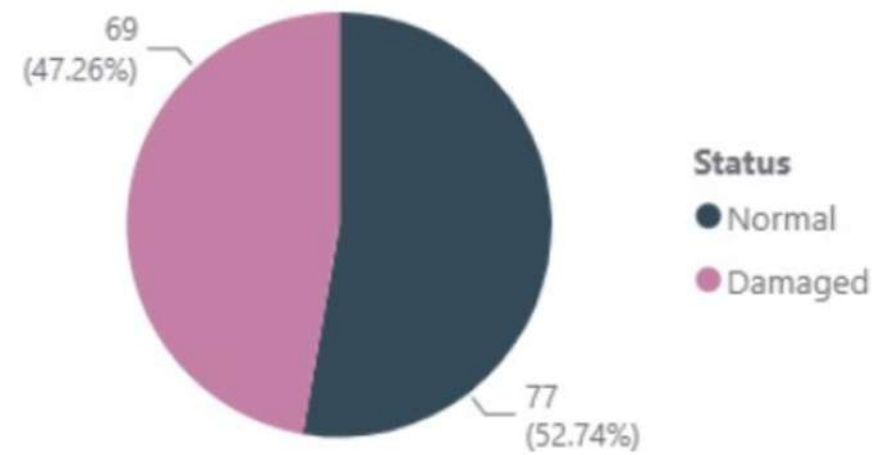
0.89

Average Confidence

Inspection Required

Conveyor Status

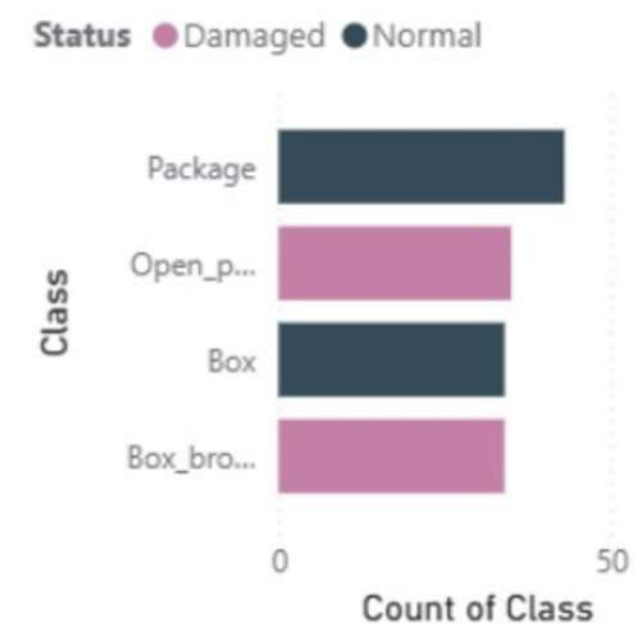
Accepted vs Rejected Parcels



Parcel Types Detected



Damage Analysis by Parcel Type



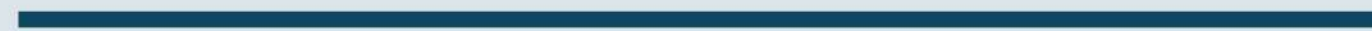
Conclusion

The Automated Parcel Damage Detection System was successfully developed using the YOLOv11 deep learning model to identify normal and damaged parcels. The dataset was annotated and augmented in Roboflow, and the final model was trained in Google Colab using GPU acceleration. The trained model was deployed in a Python inference pipeline, which automatically generated prediction results and stored them in a CSV file. These results were further visualized through an interactive Power BI dashboard for efficient monitoring and analysis. The system achieved high accuracy with 95.2% mAP@50, demonstrating its effectiveness in automating parcel inspection, reducing manual effort, and improving operational efficiency in e-commerce fulfillment centers.

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Thank you



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